

Probability Game Project: “*Lucky Draw Showdown*”

1. Introduction

Probability is used to predict how likely events are to happen, especially in games of chance. In this project, I designed a game called **Lucky Draw Showdown**, which combines concepts such as theoretical probability, combinations, independent events, and expected value.

The goal of the game is to create a fun but mathematically balanced system where players can win or lose money based on probability.

2. Game Rules

Materials:

- A bag containing **10 balls**:
 - 4 red
 - 3 blue
 - 2 green
 - 1 gold

How to Play:

1. A player pays **\$3** to play.
 2. The player randomly draws **2 balls without replacement**.
 3. The outcome depends on the combination of colors drawn.
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3. Winning Conditions & Payouts

Outcome	Description	Payout
Gold ball drawn	At least one gold ball	\$10

Two same color	(RR, BB, GG)	\$5
Two different colors	(not including gold)	\$2
No match / other cases		\$0

4. Probability Calculations

Total possible outcomes:

We use combinations since order does not matter:

$$\text{Total outcomes} = \binom{10}{2} = 45 \quad \text{Total outcomes} = \binom{21}{2} = 45$$

Case 1: Gold ball drawn

Number of ways to draw at least one gold:

- Gold + any other ball:

$$\binom{1}{1} \cdot \binom{9}{1} = 1 \cdot 9 = 9$$

Probability:

$$P(\text{gold}) = \frac{9}{45} = 0.20$$

Case 2: Two same color (excluding gold)

- Red: $\binom{4}{2} = 6$
- Blue: $\binom{3}{2} = 3$
- Green: $\binom{2}{2} = 1$

Total:

$$6 + 3 + 1 = 10$$

Probability:

$$P(\text{same color}) = \frac{10}{45} \approx 0.222$$

Case 3: Two different colors (no gold)

Total non-gold balls = 9

$$\binom{9}{2} = \frac{9!}{2!(9-2)!} = \frac{9 \times 8}{2} = 36$$

Subtract same-color cases:

$$36 - 10 = 26$$

Probability:

$$P(\text{different colors}) = \frac{26}{45} \approx 0.578$$

5. Expected Value (Important!)

We calculate the expected payout:

$$E = (0.20)(10) + (0.222)(5) + (0.578)(2) = 0.20(10) + 0.222(5) + 0.578(2) = 2 + 1.11 + 1.156 = 4.266$$

Expected payout $\approx$ \$4.27

Since players pay \$3:

$$\text{Expected profit for player} = 4.27 - 3 = +1.27$$

👉 This means the player has an advantage (not good for the game host).

6. Adjusted Fair Game

To make the game fair, we can:

- Lower payouts OR
- Increase cost to play

Example adjustment:

- Keep payouts the same

- Change cost to play = **\$4.25**

Now:

Expected profit ≈ 0

7. Conclusion

The game **Lucky Draw Showdown** demonstrates how probability can be applied to real-life situations like gambling and game design. Using combinations and expected value, we found that the original version of the game favored the player. By adjusting the entry cost, we made the game fair.

This project shows the importance of probability in ensuring games are balanced and not biased toward either the player or the organizer.